



EUROPEAN MICROSOFT FABRIC S&L Community Conference

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From Notebook to Bedside: Building a Clinical Risk Prediction Pipeline in Fabric



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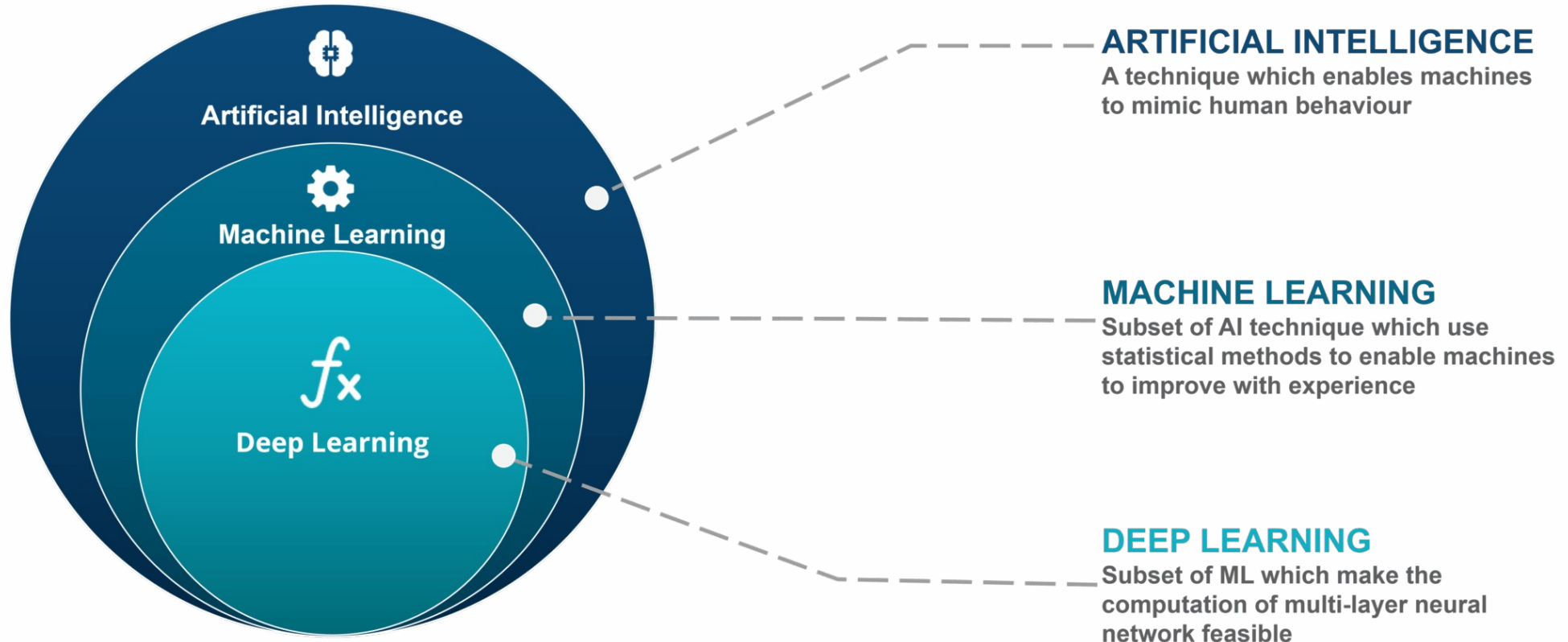
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Agenda

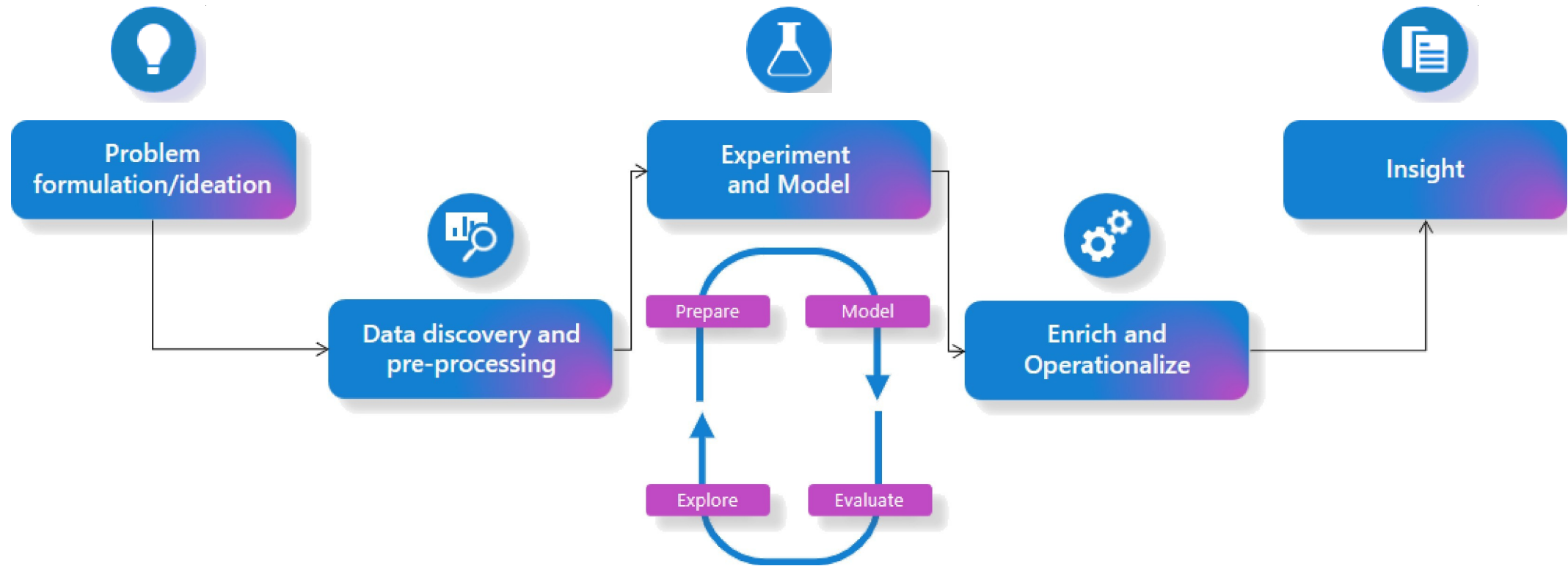
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Fabric Data Science & Machine Learning

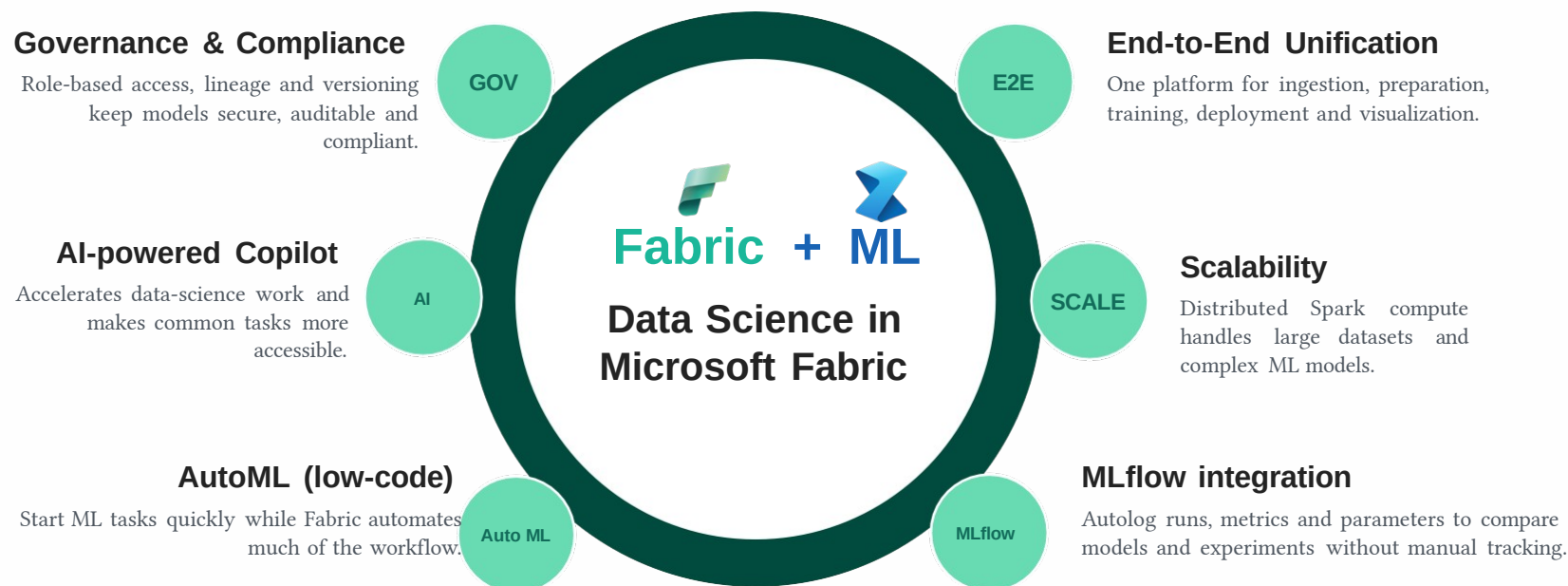
ML as a subset of AI



The data science process



Microsoft Fabric as a platform for Data Science

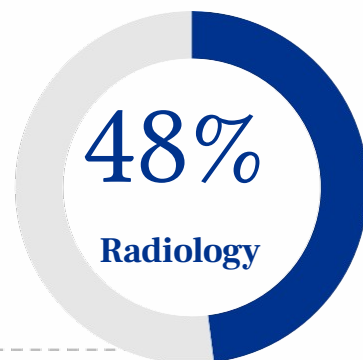


Prediction Models in Healthcare

AI and ML is being embedded in EU healthcare systems

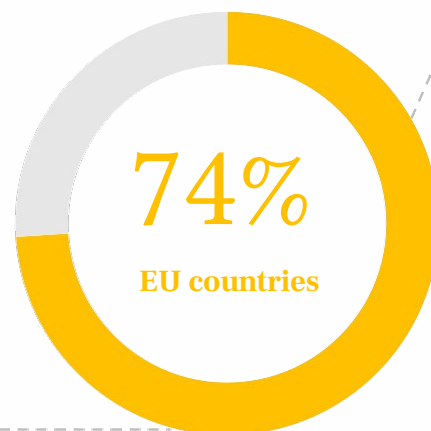
Radiology reality

48% of 572 European radiology respondents used AI; another 25% planned adoption.



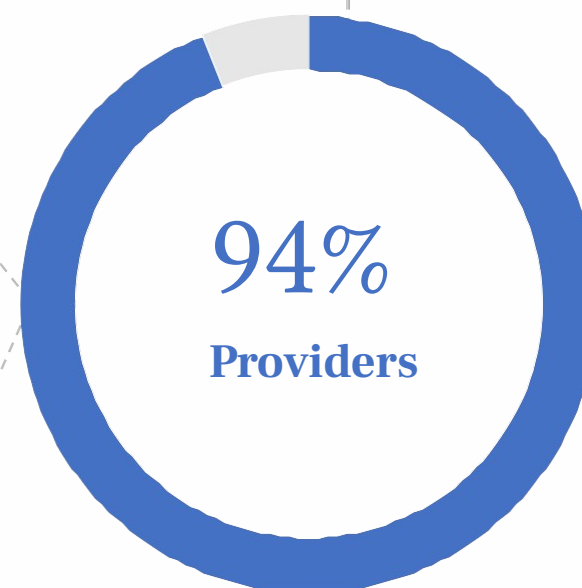
Diagnostics spread

Nearly 3 in 4 EU countries report AI-assisted diagnostics in clinical settings.



Provider pipeline

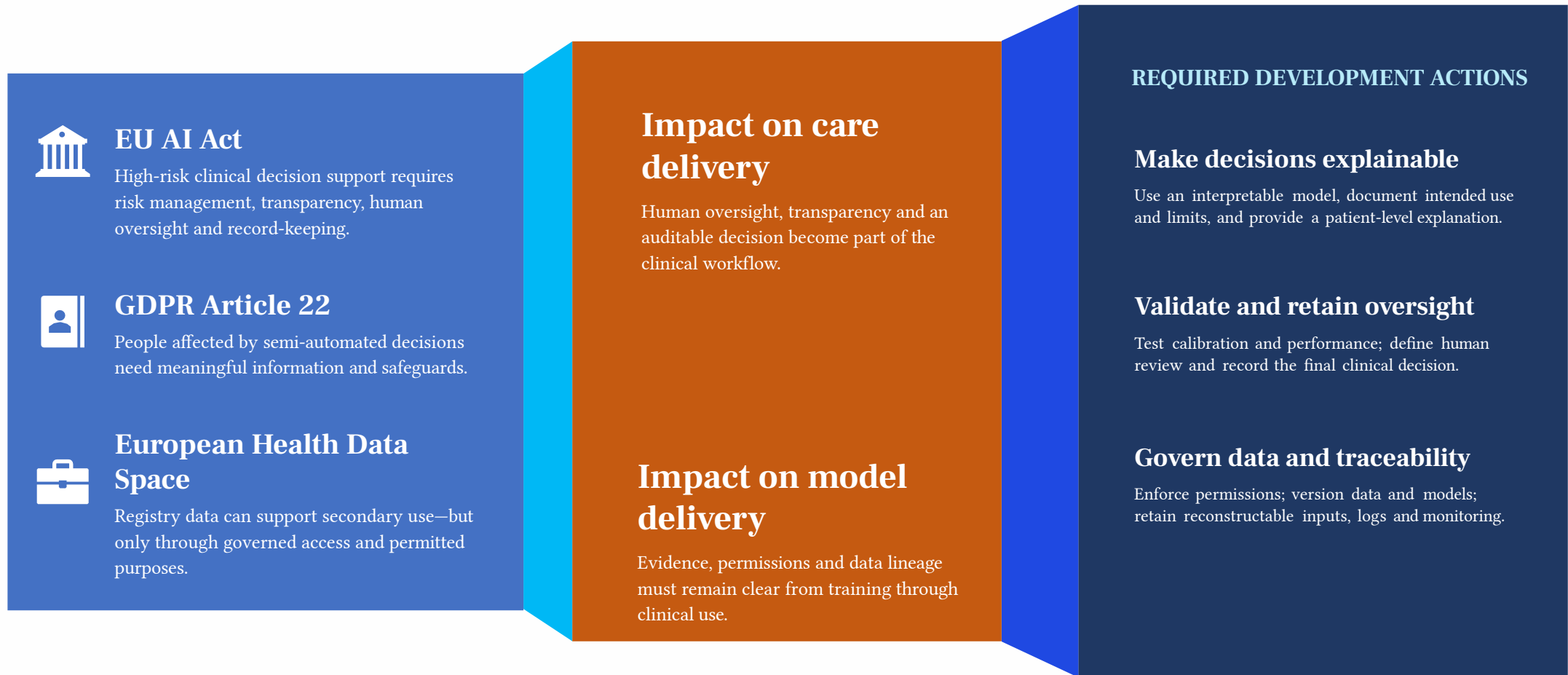
94% of surveyed EU healthcare providers already use AI solutions or plan to adopt them.



Measurement caveat: no harmonized EU hospital-level adoption ranking exists; these indicators use different denominators and are not directly comparable.

Sources: WHO/Europe EU readiness survey (2024–25); European Commission Digital Health Observatory (2026); Zanardo et al., Insights into Imaging (2024).

However, EU rules turn healthcare ML into a governed clinical process



Yet not all institutions are prepared...

Box E3. Legal and regulatory landscape for AI in health: key findings

7%

(2 of 27) have issued guidance for manufacturers and users on applying existing liability regimes to AI



48%

(13 of 27) have implemented data accountability practices as a minimum standard for AI governance in health



56%

(15 of 27) have plans to align legal liability standards with upcoming EU legislation



22%

(6 of 27) have issued practical guidance on ethics by design for AI systems in health



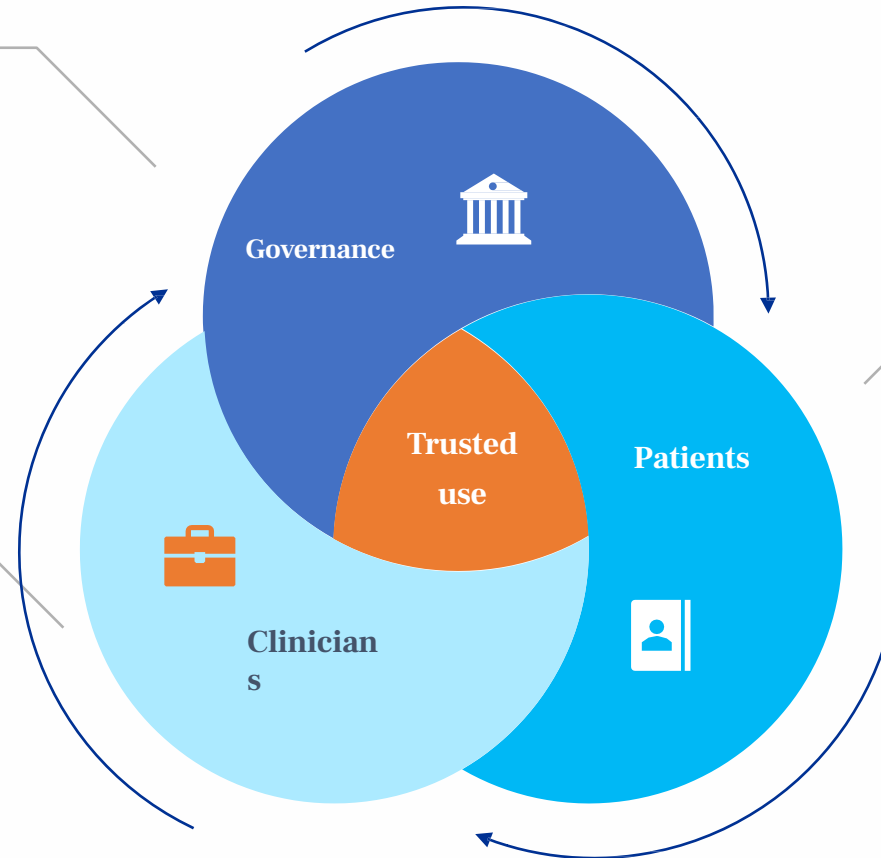
Three audiences face different obligations

Regulators & governance

Need auditable lineage, versioning, monitoring and a recorded human decision for every use.

Clinicians

Need an interpretable, calibrated score in an approved workflow—while retaining final accountability.



Patients

Need a clear explanation of the score, its uncertainty and how it influenced—or can be challenged in—care.

Fabric Healthcare Data Solutions

Healthcare data solutions

Improve the efficiency and value of your healthcare data

These healthcare data solutions help you unify disparate data sources running on Fabric. Each capability contains a unique set of tools to help you prepare the data for ingestion, transformation and harmonization for advanced analytics and AI modeling.

Deploy the relevant solutions to your workspace to get started on your healthcare data transformation goals.



Setup your solution to get started

The setup module will set up your healthcare data estate. You can select, configure and deploy capabilities such as data connectors and sample data in a guided, step-by-step process.

[Setup your solution](#)

Solution capabilities and sample data

Select each capability to learn about it and build your data transformation solution.



Sample data

Test your healthcare data solutions using this sample data.



Azure health data services - data export

Export data from the FHIR service in Azure Health Data Services to OneLake in Fabric.



CMS Claims data transformations

Bring your CMS CCLF (Claim and Claim linefeed) data to OneLake.



DICOM data transformation

Bring your DICOM (Digital Imaging and Communications in Medicine) imaging data to OneLake.



OMOP transformations

Prepare data for standardized analytics through OMOP (Observational Medical Outcomes Partnership) open community standards.



SDOH datasets - transformations

Bring social determinants of health (SDOH) datasets to the OneLake to unify it with core healthcare data domains such as clinical and claims.



Dynamics 365 Customer Insights - data preparation

Connect Dynamics 365 Customer Insights to your OneLake on Fabric for creating patient or member lists for your outreach.



Patient outreach analytics

Bring marketing and analytics data from Dynamics 365 to OneLake. Harmonize this data with your patient data and leverage our Power BI templates to improve patient engagement.



Care management analytics

An analytical template that combines data from various key modalities of healthcare data solutions to provide insights and make data driven decisions.



DAX Copilot data transformation

Prepare DAX Copilot data for analytics



Orchestrate multimodal AI insights (Public Preview)

Enable on-demand generation and integration of AI enrichments from our 1P healthcare AI models and APIs.

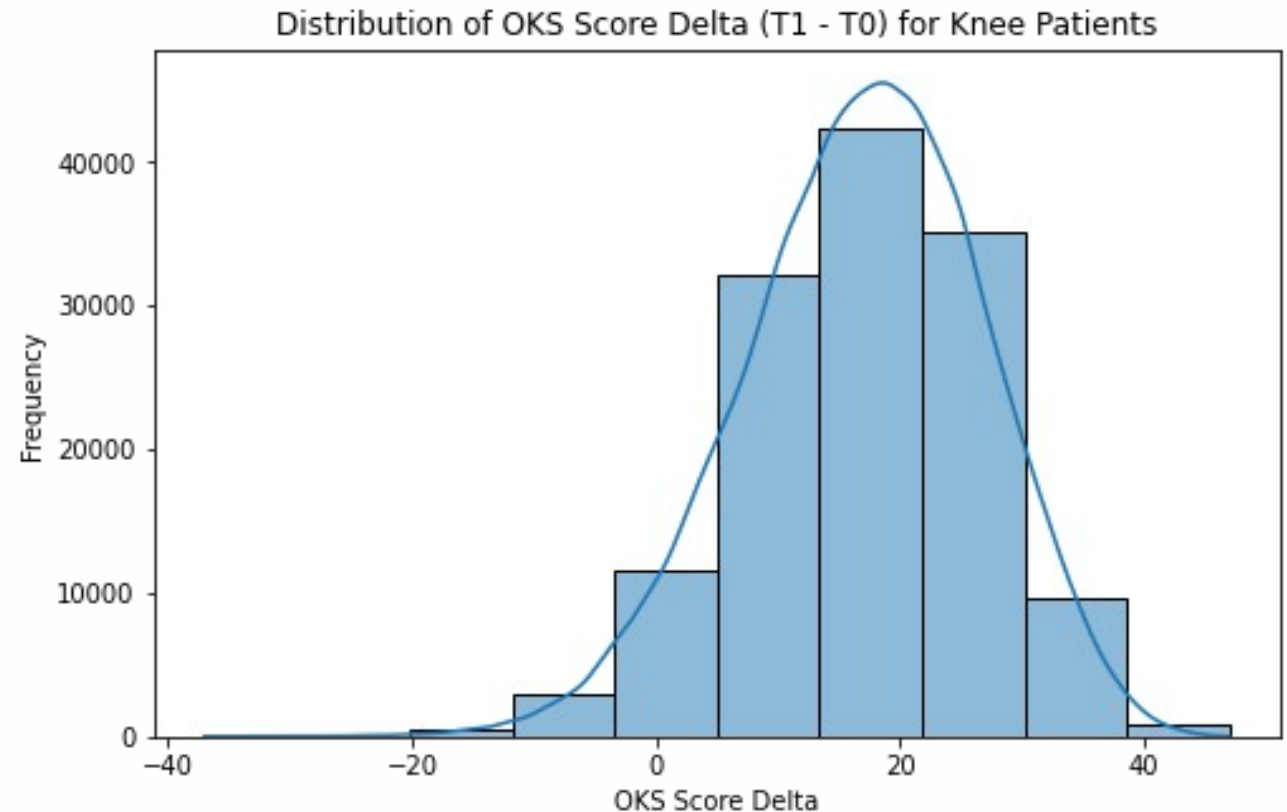
The case: knee replacement on NHS PROMs

NHS PROMs: the prediction starts before surgery

A pre-operative cohort, a six-month outcome, and a clinically meaningful threshold

One row per episode

Pre-operative OKS, EQ-5D-3L, comorbidities, mobility and provider context form the feature set. The outcome arrives when the patient returns the questionnaire six months later.



The knee cohort concentrates around a positive change, which makes the “poor outcome” definition a clinical choice rather than a generic label.

What counts as a poor outcome?

Oxford Knee Score runs from 0 to 48. A 7-point improvement is used as the minimum clinically important difference.

OKS at six months - OKS before surgery

< 7 points
Poor outcome

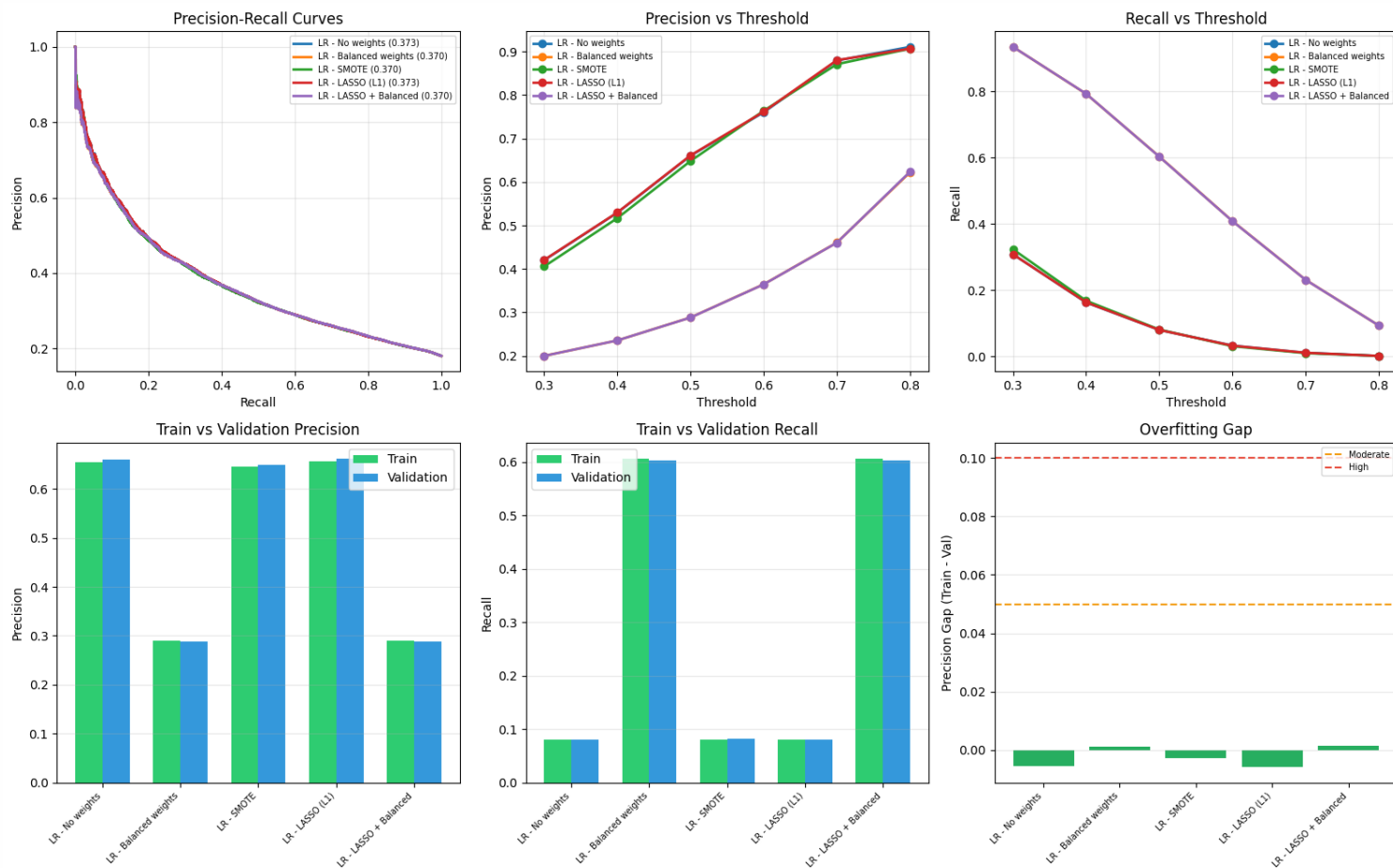
≥ 7 points
Meaningful improvement

The model only receives information available before the decision to operate.

That keeps the score suitable for a pre-operative conversation. It does not diagnose a patient or decide treatment.

Exploration drove the modelling choices

The notebook outputs keep the choices inspectable instead of hiding them inside training code.



Three questions before training

How common is the poor-outcome class?

Which variables are missing by design?

Which threshold fits the capacity for follow-up?

Train and test split

The preprocessing pipeline is fitted on training data only. The held-out set stays untouched until evaluation.

01

Stratified split

Preserve the poor-outcome rate in both sets

02

Fit transforms

Imputation and encoding learn from training rows

03

Evaluate once

Use the held-out cohort for model comparison

Leakage prevention is part of the model design, not a last-minute validation check.

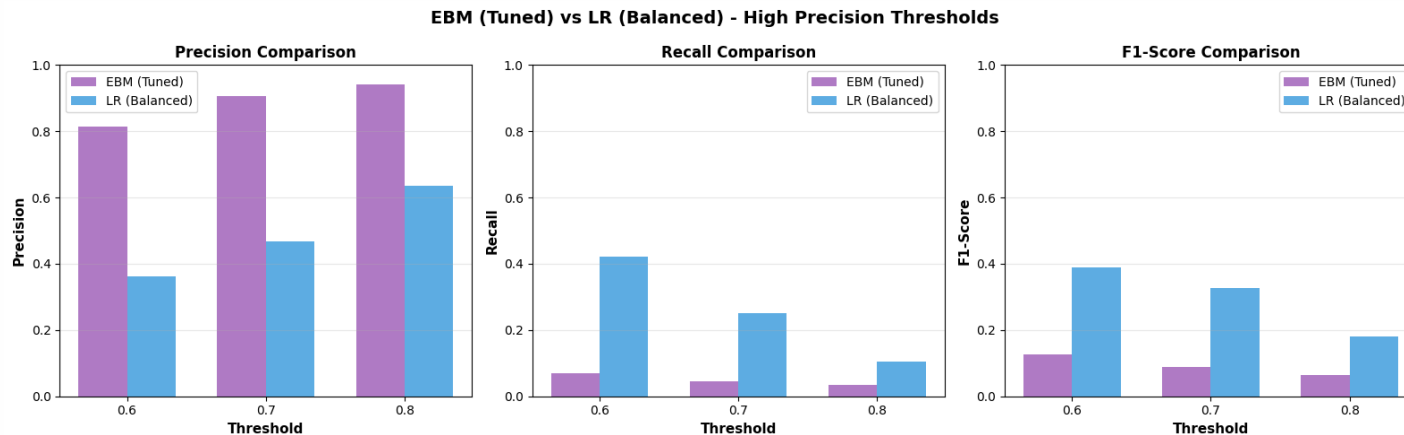
Model selection favoured an explainable model

The experiment compared linear baselines, tree models and an Explainable Boosting Machine.

Why the EBM?

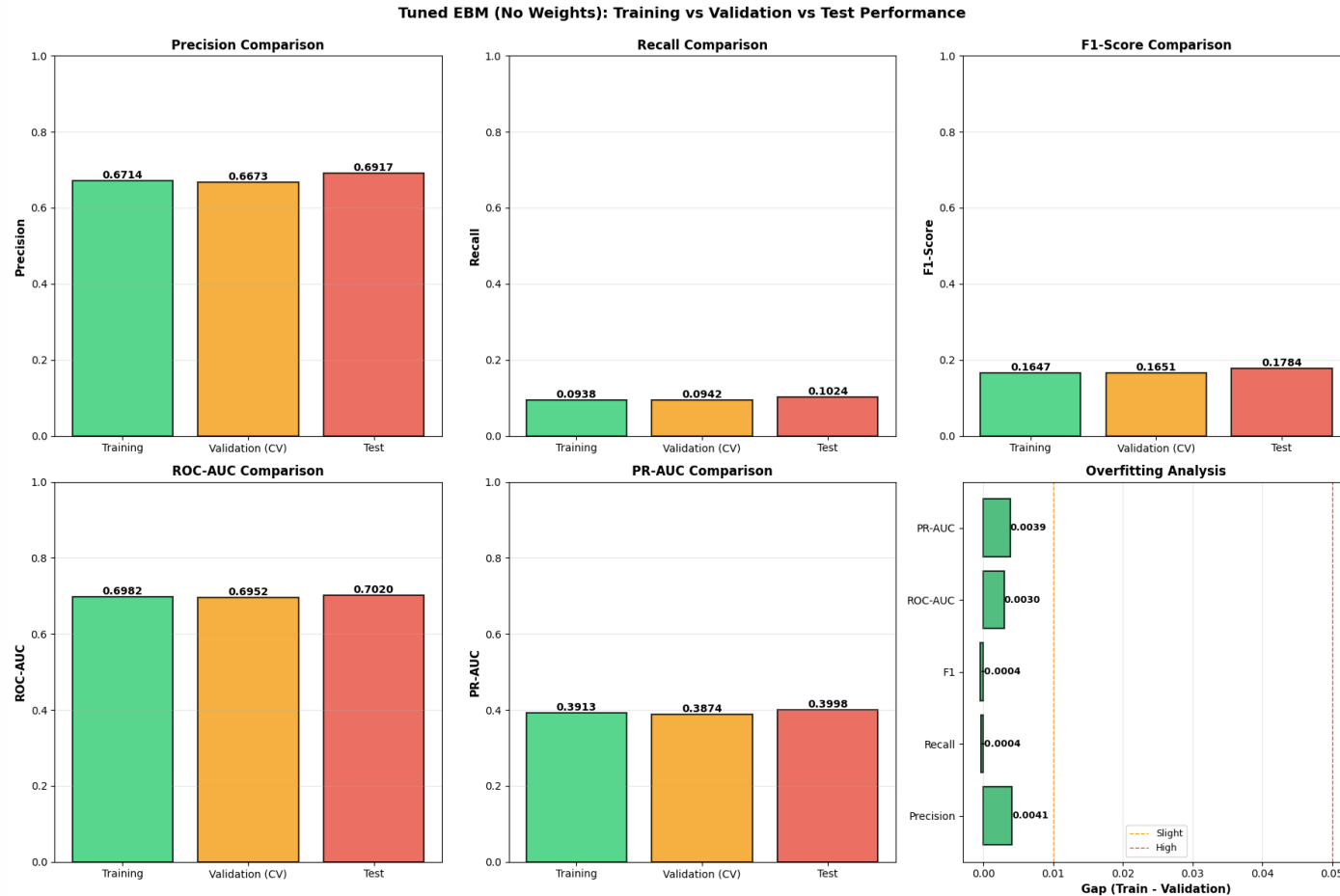
It provides a calibrated risk score and a direct explanation of how each input shifts risk.

That lets clinicians inspect both the model-wide pattern and the individual prediction.



Performance depends on the threshold

Precision rises as the flagging threshold rises. Recall falls. The operating point is a care-pathway decision.



A score is not a decision

The model estimates risk. The service chooses the threshold based on how many patients it can support and the cost of missed cases.

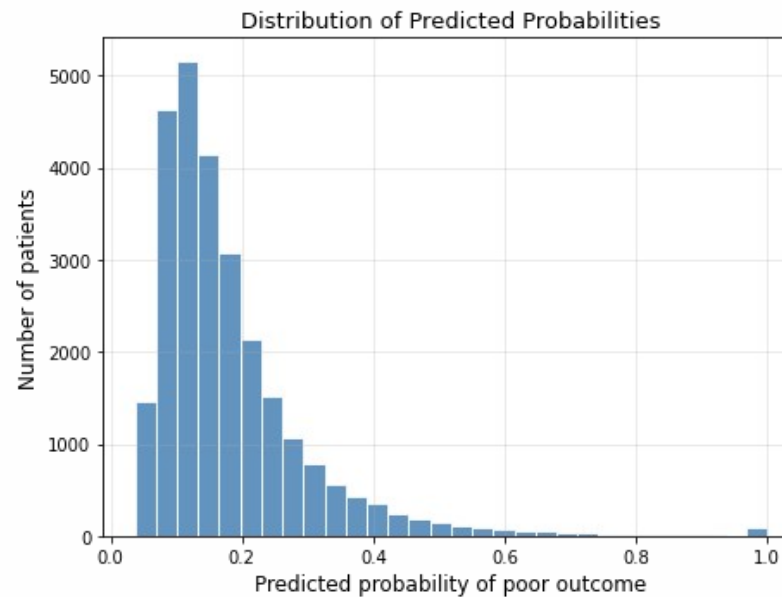
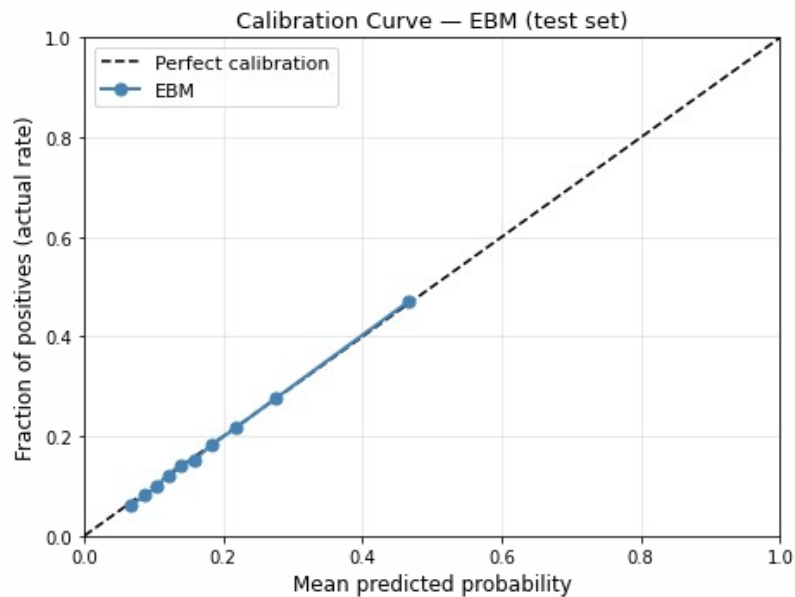
Threshold Policy choice

A model a clinician can argue with

Global behaviour, local explanations, and calibration

Calibration makes a risk score interpretable

When the model estimates 34%, the observed poor-outcome rate should be close to 34% for similar patients.

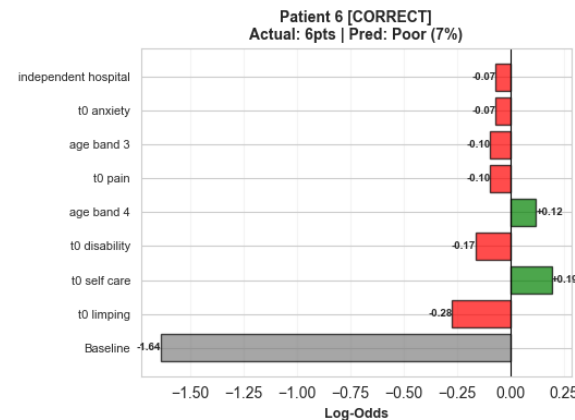
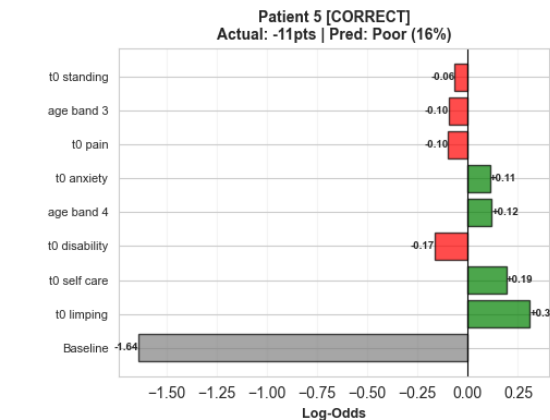
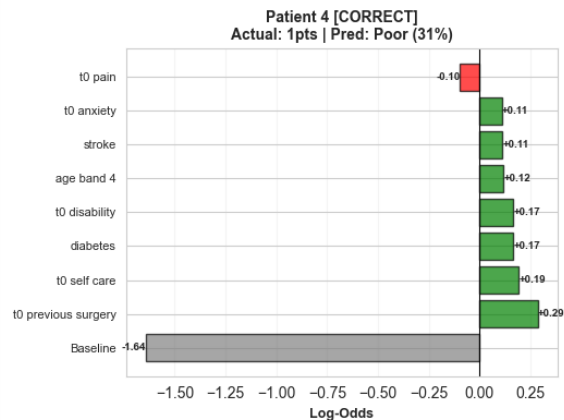
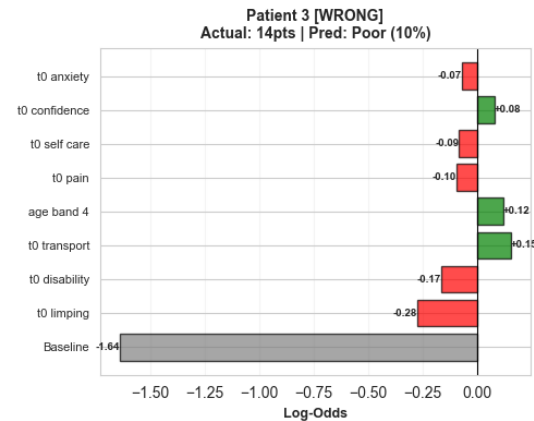
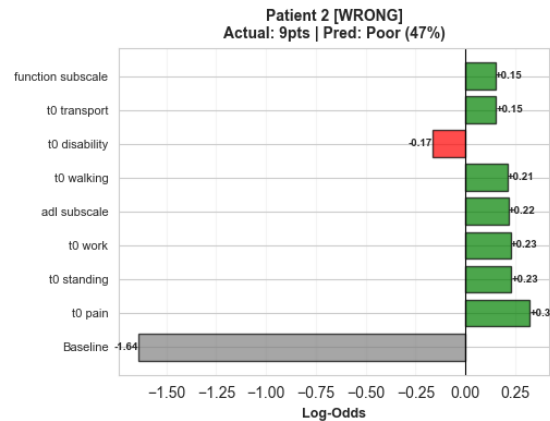
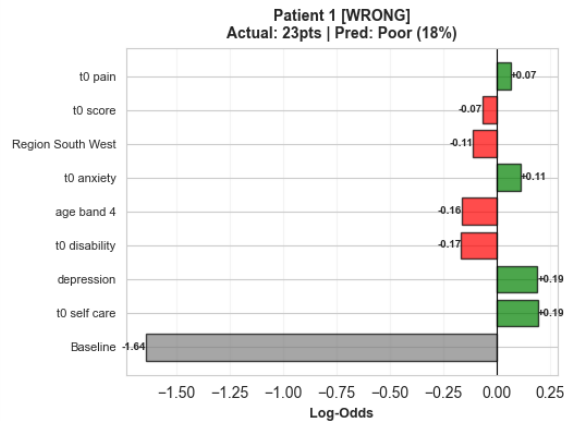


A probability needs a reality check

The curve follows the diagonal closely across the observed probability range. The distribution shows most patients below high-risk territory.

Each patient receives a local explanation

The model exposes signed feature contributions instead of a black-box score.



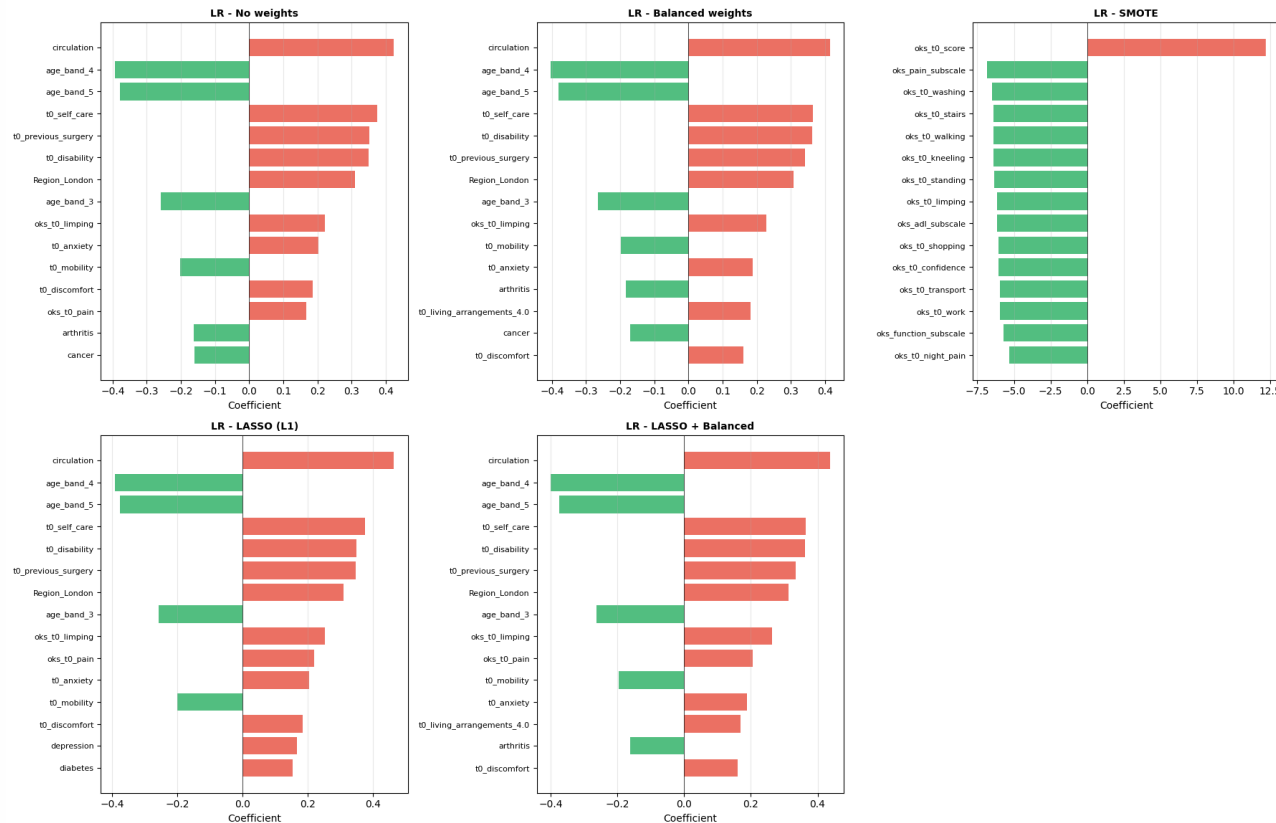
What changes the conversation

The clinician can see which pre-operative factors push risk up or down, then decide whether the result fits the patient's context.

Global patterns remain inspectable

A transparent model supports a discussion about patterns across the cohort as well as a discussion about one patient.

Feature Importance (Top 15) - Green: Positive, Red: Negative



Two views of the same model

Global view
Which pre-operative features matter most across the cohort?

Local view
Why did this patient receive this score?

A score must survive an audit

The model record joins the data version, feature set, threshold and model version to the clinical decision.



“Which input created this score, and who acted on it?”

From experiment to registered model

Fabric keeps the training lineage and the deployable model in the same governed estate.

ML experiment

Each training run records metrics and explicit tags such as data version and interpretability. ➤

Registered model

The selected calibrated EBM receives a version and a champion alias before it ➤ reaches scoring.

Scored cohort

Batch scoring writes risk and explanation tables that the clinician app reads.

No model file needs to leave the estate to reach the workflow.

Delivery inside the governed estate

Versioned model · scored cohort · clinician workflow · access policy

The score carries its own evidence

The endpoint response brings risk, threshold, top contributions and lineage metadata together.

```
{  
  "episode_id": "...",  
  "risk": 0.34,  
  "threshold": 0.50,  
  "flagged": false,  
  "contributions": [ ... ],  
  "model_version": "champion",  
  "gold_table_version": 7  
}
```

Why this score? Designed for
the endpoint response:

“Why this score?”

“Which model and data version produced
it?”

“Was the patient flagged at the active
threshold?”

The clinician app turns a score into a review task

The app uses the scored cohort and explanation tables, without exposing a generic model interface to a clinician.

Patient review

Risk band
High

Why?
Reduced mobility
Baseline pain
Prior surgery

Record decision

The workflow supports judgement, not automation

A model score identifies a patient for review.
The clinician sees the explanation, adds context and records the decision. The app does not make the care decision.

Row-level security sits beside the data

The access rule is evaluated by Data API Builder before a patient row reaches the app.

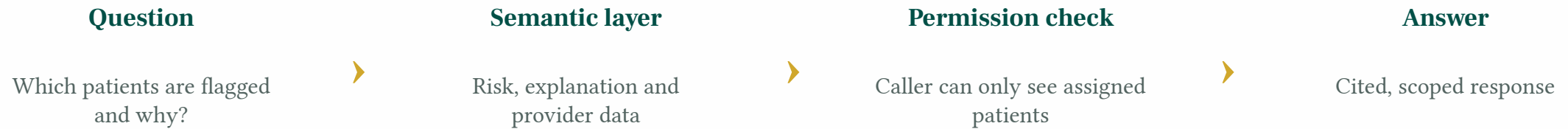
```
policy: (claims, item) =>  
  claims.sub  
    .eq(item.assignedClinicianId)  
    .or(  
      claims.email.eq(  
        item.assignedClinicianEmail  
      )  
    )  
  )
```

No client-side filter

Filtering in the frontend only hides rows in one screen. The data policy protects every request, including deep links and API calls.

A grounded answer, or none

The agent can answer from the governed model outputs and provider context, with the same permission boundary.



The answer should remain traceable to the scored tables and the caller's permissions.

The missing layer is delivery with evidence

Clinician

Risk, explanation and a recorded decision

Patient

A clear, calibrated probability

Regulator

Lineage, model version and access policy

Data, analytics and decision remain in one governed place.

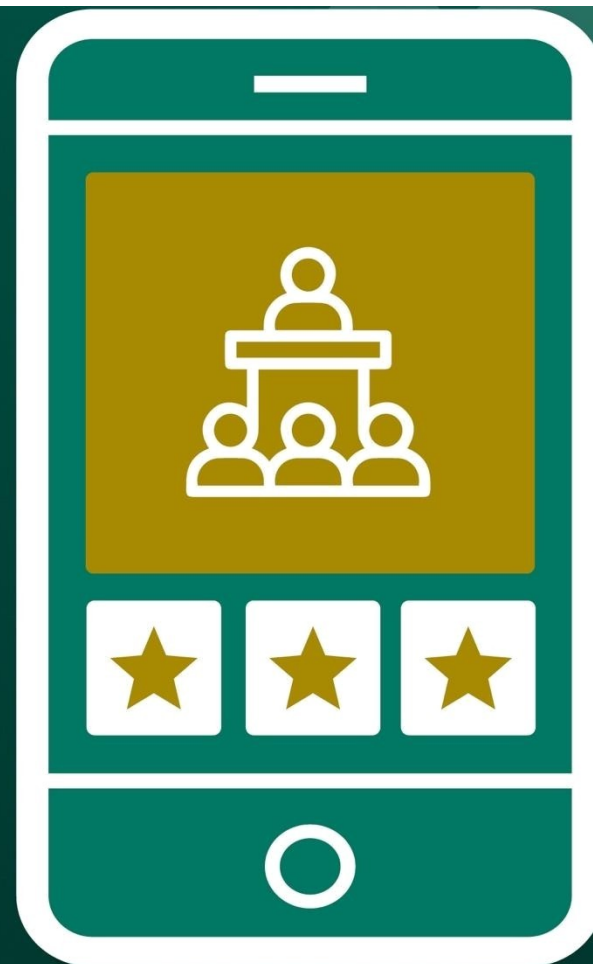
What transfers to the next project

The reusable work is not only the model. It is the governed path from data to decision.

- 1** **Version the inputs** Record the Delta table and feature-set version with each run
- 2** **Make explanations first-class** Store local explanations beside the score
- 3** **Treat thresholds as policy** Connect flagging volume to the care pathway
- 4** **Enforce access in the data layer** Do not rely on a frontend filter
- 5** **Enforce a stable pathway** Pre-run notebooks and offline app mode for the demo



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